classmate
Date
Page 1

॥ ॐ नमः शिवाय ॥

5 June 2025 – Thursday – 10:35 AM – Softvan – 3rd floor

Cloud subject —— 2 days basic
15 days internship
- Mostly practical

Capex & Opex Operational expenditure
Reason for IT Boom

Capital Expenditure

Zomato use reverse

Cloud is IMP for AI for hosting AI.

Vibe coding — AI led code.

AI can't create pipelines / Architecture

||ॐ नमः शिवाय||

||ॐ नमो भगवते वासुदेवाय नमः||

6 June 2025 — Friday — 9:30AM — Dahana Office

Cloud Computing

BOQ → Bill of Quantity — Individual pricing of each component.

Accessing data from anywhere in the world at any time in real time dynamic space.

On demand computer.

Pehle difference ye tha ki computers ka size fix tha aaj dynamic kar diya hai.

Instagram Story — Bar Table —

||ॐ नमः शिवाय||

||ॐ नमो भगवते वासुदेवाय नमः||

7 June 2025 – Saturday – 9:50 AM – Sahana Office

LendingKart – financial lending platform

DOCOMO – Airtel of Japan

Choice Groups – Hayat, Lemon Tree –

Cisco LLM based on AWS

PGA Tour – Golf on AWS

NASA –

Capital One – Billion dollar bank

Caratlane Tata Premium diamond

* Edge Location

What is Docker? – Compute in backend

Services
1. Compute
2. Storage
3. Database
4. Network

Computer Networks IMP Subject.

- Infrastructure as a service (IAAS)
- Platform as a service (PAAS)
- Software as a service (SAAS)

→ Private Cloud — Office
→ Public Cloud — Amazon, Google
→ Hybrid Cloud — Banks, Flipkart (Purchase Bank)

||ॐ नमः शिवाय||

||ॐ नमो भगवते वासुदेवाय नमः||

9 June 2025 – Monday – Sahara Office – 9:38 AM

||ॐ नमो भगवते वासुदेवाय नमः||

||ॐ नमः शिवाय||

10 June 2025 — Tuesday — Sahara office — 9:37 AM

EC2 → Elastic computing 2

VPC → Virtual Private Cloud

RDP → Remote Desktop Protocol

IOPS → Input output Operations per Second

Temporary files are stored here, which are going to get deleted afterwards

for Main storage we use S3

default memory is 30 gb for C Drive

1) EC2 → Instance → Launch an instance (Key value pair required)
2) Instance ID
3) Connection Type : connect using RDP client
4) download Remote Desktop file (file is encrypted)
5) Get password (upload RDP file)

for Mac RDP tool.

Instance mein web application server download krna hoga.

Application hosting → Web App Server

Apache Tomcat & JDK download in Server

Tomcat default port → 8080

SSH file for Ubuntu/Mac

Startup batch file for Windows.

8080 port allow from instance at AWS

Instance → Security → Sgroup → Inbound Rule

AddIP → Add custom host Server

॥ ॐ नमः शिवाय ॥

॥ ॐ नमो भगवते वासुदेवाय नमः ॥

CAI – 1 July 2025 – Tuesday – EB-823 – 8:30 AM

EC2 dekhna hai, Lambda

Networking
Processor
Server less architecture.

- AWS recognition → C++ backend
  Sentiment analysis, behaviour analysis.

॥ ॐ नमो भगवते वासुदेवाय नमः ॥

॥ ॐ नमः शिवाय ॥

DLCV – 2 July 2025 – Wednesday – EB-823 – 8:43 AM

MLP → Multilayered Perceptron

NVR – Network
DVR – Digital

Stream → rtsp = real time Stream protocol.

YOLO in DLCV | Apple Beacon

Transfer Learning

Quantisation: – reducing parameters

AWS

Page No.
DATE: / /

How AWS work.?

1. global data center (Regions & Availabilityzone)
2. Core infrastructure Backbone
3. Secure cloud network. with Amazon VPC.
4. Compute and Storage Services
5. Auto Scaling and load Balancing
6. User friendly management Console
7. Built for high Availablity and fault tolerance

* Amzone EC2 ⇒ provide Scalable virtual Server for running application.
* Amzone S3 ⇒ offer highly durable and Scalable cloud Storage.
* Auto Scaling :- Automatically adjust the no. of EC2 instance based on demand.
* Elastic load Balancing (ELB) :- Distributes traffic across multiple Server for better performance and uptime.

Page No.
DATE: / /

Q. Cloud Computing

- Huge space online.
- collection of server.
- orchestration.

⇒ Use Case

- manufacturing organization → Expand Business
- media Company → Expand Business
- architecture Consulting → Compute Rendering for prototyping
- large Enterprise → Deliver training

⇒ Advantages of AWS

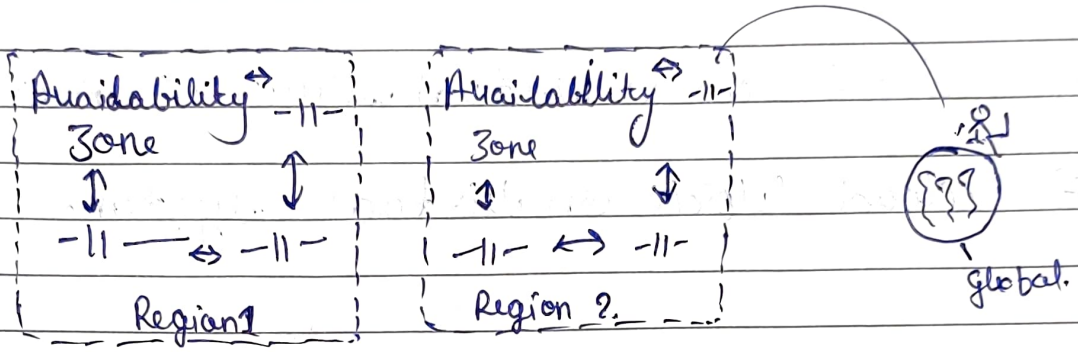
- flexibility.
- Cost Effective.
- Scalability.
- Security.

Page No.
DATE: / /

⇒ ~~AWS global~~

⇒ AWS global infrastructure.

- divide into Categories: Region and Availability Zone.



⇒ Domain of AWS: imp Service

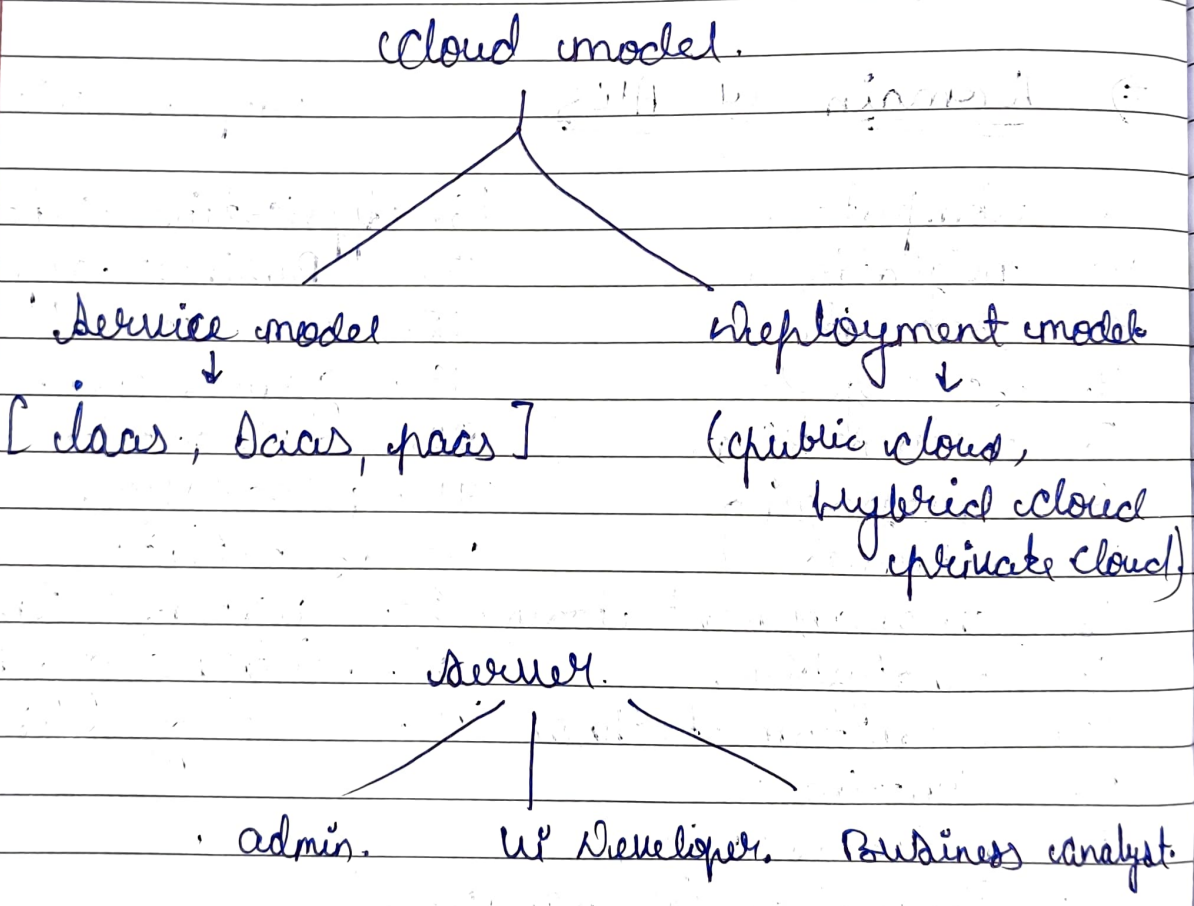
1. Compute.
2. migration.
3. Security.
4. Storage.
5. networking.
6. messaging
7. database.
8. management tools.
9. Analytics.
10. quantum techno.
11. Containers
12. media services
13. Cloud financial management
14. frontend - Web. & mobile.
15. ML.
16. Application integration
17. Business application
18. Dev. tool.
19. game. dev.
20. Robotics
21. Blockchain
22. Satellite
23. Customer Enablement
24. internet of thing
25. End of User compute

Page No.
DATE: / /

* Deployment model.
  - cloud based
  - on-premises.
  - Hybrid.

⇒ EC2 ⇒ is a virtual server in Aws cloud.

⇒ cloud work on virtual system.



IAM - identity access management. SNS
- is web services.
→ control the user access.

Page No.
DATE: / / Cloud watch

⇒ Resource monitored By Cloud Watch.

1. Virtual instance (EC2).
2. Data Base (AWS RDS).
3. Data stored in amzone S3.
4. Elastic load Balancers.
5. other AWS Resources.

⇒ How Amzon CloudWatch Work?

Amzone CloudWatch

⇒ Collect
- metrics & logs are collected

⇒ monitor
- visualize metrics & log.

⇒ act
- automated responce

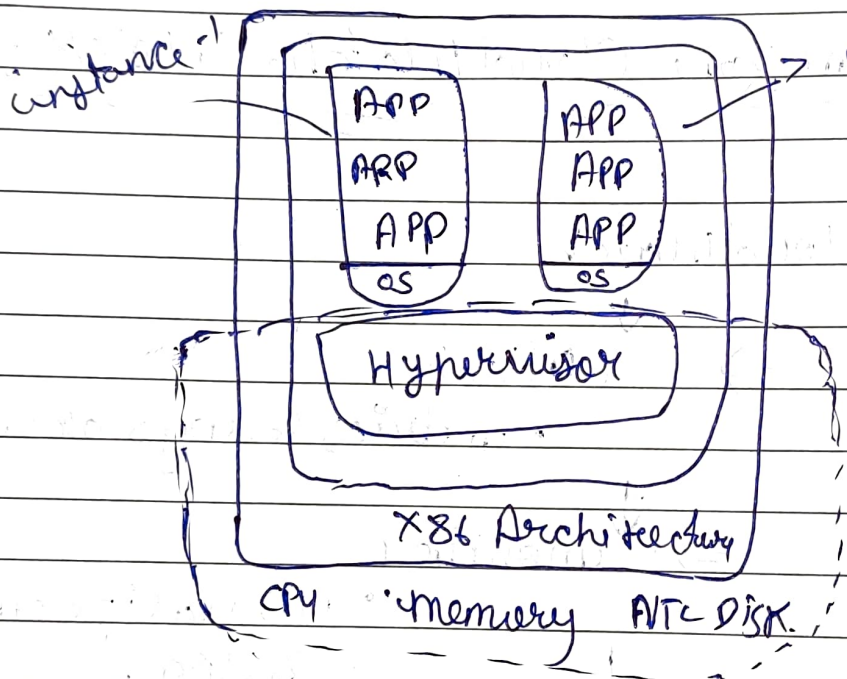
⇒ analyse
- Real time analysis

- Health moni.
- Application moni.
- System wide visibility
- Resource optimizatio

→ Vm Ware create a product name is "Hypervisor".

Let Explain

Hardware like laptop. तो उसके अंदर multiple os data. है तो उसके लिए Hypervisor data पड़ेगा।



→ This responsibility of AWS manage AWS ⇒ Paas (picture as a Service)

day20.

Page No.
DATE: / /

⇒ What is AMi?

Amazone machine image.

⇒ an Ami is a pre-configured template of a Virtual ~~mac~~ Environment in AWS.

- it include the os, application server, application, and other configuration needed to deploy a virtual machine.

- Use Case.

⇒ Rapid instance deployment of EC2 instance
⇒ load Balancing and auto scaling
⇒ Environment Cloning
⇒ Disaster Recovery.

---

public ip

- ISP (AWS)
- Routable on internet

private IP

A - 10.0.0.0 - 10.255.255.255
B 192.16.0.0 - 172.31.255.255
C - 198.168.0.0 - 192.168.255.255

- not Routable on internet

⇒ S3 ⇒ S3 like work as a storage.

MFA

⇒ Cloud Computing : Cloud Computing is the on demand delivery service. including

→ Server.
→ Storage
→ data base
→ analytics
→ intelligence over the network.

EC2

Page No.
DATE: / /

⇒ What is EC2 ?

- EC2 like a virtual Computer server that run on cloud.
- you can install software, host website, Run Code and do everything like on your lap or pc.

⇒ Why :

- you don't need to buy & manage physical machi
- you can scale it up and down anytime based on your need.
- it run 24/7 if you want, perfect for website application etc.

for Mac RDP tool.

Inst.

Page No.

DATE: / /

S3

⇒ What is S3?

- S3 like a cloud hard disk to store any file photo, video, PDF, backup Etc.
- you upload and download file via the internet

⇒ Why?

- it cheap, durable and can store unlimited data
- good for backup, media file or data used by app.
- you can make file private or public (like for a website)

IAM (identity access management.

Page No.
DATE: / /

What is IAM ?

- IAM is used to manage who can access your Aws resource and what they can do.
- like giving key with rules to your Aws House.

- Why.
- you want to give limited to user. Eg. developers, admins Etc.
- it keep your Aws account secure.

Page No.
DATE: / /

lambda

⇒ What is lambda.

- lambda lets you run your code without creating server.
- just upload your code and AWS runs it only when need.

⇒ Why?

- no need to manage server.
- you only pay when your code run
- good for automation, quick task, API.

=

- Code ko trigger kar sakte ho: kisi kam ho ne par automatically.

Example

- we have upload a image on website..
- then now this image store in secure folder automatically.
- so we use lambda: now lambda create ek small version (thumbnail) is create automati.

Saas	Iaas	Paas?
1. Software as a Service • Ek client ko apne business aage ya promote kar ne keliye ek app ya website bannani hai to. ish client pura saas ko dedeta kam like 0 to 100 app complete saas developing or Engineer like, frontend Backend, database cloud.	1. infrastructure as a Service. "Apne puri website ya capp, bana diyo ab usko host karna chai toh usko liye storage chaiye, network chaiye, toh apko usko liye pura infrastructure chaiye toh. Vo Cloud provide kar dega.	• platform as a Service • provide like database, Storage, and os Server. for help to develop or create appli. or website so paas provide as a platform means • ki apko koi bhi cheeze ki zroorat chai toh ham provide karege.

Aws Task

Page No.
DATE: / /

- ~~Aws~~ task

- Create instance from Ami ✓
- Create Ami from instance. ✓
- Apache server to host Web Site.